

AI & ML Internship - Task 16: Chatbot (Rule-Based + ML Hybrid)

1. What is a Chatbot?

A chatbot is a software application that interacts with users and provides automated responses.

2. What is NLP?

Natural Language Processing enables computers to understand and process human language.

3. What is Tokenization?

Tokenization is splitting text into smaller units called tokens.

4. What is TF-IDF?

TF-IDF measures the importance of words in a document.

5. What is Intent Classification?

Identifying the purpose of a user's message.

6. What is a Rule-Based Chatbot?

A chatbot that responds using predefined rules and keywords.

7. What is an ML-Based Chatbot?

A chatbot that predicts user intent using machine learning.

8. What is Naive Bayes?

A probabilistic algorithm used for text classification.

9. What is SVM?

A supervised learning algorithm used for classification tasks.

10. What is a Hybrid Chatbot?

A chatbot combining rule-based logic and machine learning.

11. Why use TF-IDF in chatbots?

To convert text into numerical features for ML models.

12. What is Text Preprocessing?

Cleaning and preparing text before training models.

13. What is Stemming?

Reducing words to their root form.

14. What is Lemmatization?

Converting words to their meaningful base form.

15. What are Stop Words?

Common words removed to improve NLP performance.

16. How does a Hybrid Bot differ from a Generative Bot?

Hybrid bots use rules and ML; generative bots create new responses.

17. Applications of Chatbots?

Customer support, healthcare, banking, education, and e-commerce.

18. Advantages of Hybrid Chatbots?

Better accuracy, flexibility, and faster responses.

19. Challenges in Chatbot Development?

Intent detection, context handling, and data quality.

20. Final Outcome of this Task?

Hands-on experience in NLP and chatbot development.